

3.1.8 Thermodynamics (A-level only)

The further study of thermodynamics builds on the Energetics section and is important in understanding the stability of compounds and why chemical reactions occur. Enthalpy change is linked with entropy change enabling the free-energy change to be calculated.

3.1.8.1 Born–Haber cycles (A-level only)

Content	Opportunities for skills development
Lattice enthalpy can be defined as either enthalpy of lattice dissociation or enthalpy of lattice formation. Born–Haber cycles are used to calculate lattice enthalpies using the following data: • enthalpy of formation • ionisation energy • enthalpy of atomisation • bond enthalpy • electron affinity. Students should be able to: • define each of the above terms and lattice enthalpy • construct Born–Haber cycles to calculate lattice enthalpies using these enthalpy changes • construct Born–Haber cycles to calculate one of the other enthalpy changes • compare lattice enthalpies from Born–Haber cycles with those from calculations based on a perfect ionic model to provide evidence for covalent character in ionic compounds. Cycles are used to calculate enthalpies of solution for ionic compounds from lattice enthalpies and enthalpies of hydration. Students should be able to: • define the term enthalpy of hydration • perform calculations of an enthalpy change using these cycles.	

3.1.8.2 Gibbs free-energy change, ΔG , and entropy change, ΔS (A-level only)

Content	Opportunities for skills development
ΔH, whilst important, is not sufficient to explain feasible change. The concept of increasing disorder (entropy change, ΔS). ΔS accounts for the above deficiency, illustrated by physical changes and chemical changes. The balance between entropy and enthalpy determines the feasibility of a reaction given by the relationship: $\Delta G = \Delta H - T\Delta S$ (derivation not required). For a reaction to be feasible, the value of ΔG must be zero or negative. Students should be able to: - calculate entropy changes from absolute entropy values - use the relationship $\Delta G = \Delta H - T\Delta S$ to determine how ΔG varies with temperature - use the relationship $\Delta G = \Delta H - T\Delta S$ to determine the temperature at which a reaction becomes feasible.	AT a, b and k PS 3.2 Students could be asked to find ΔS for vaporization of water using a kettle. MS 2.2, 2.3 and 2.4 Students rearrange the equation $\Delta G = \Delta H - T\Delta S$ to find unknown values. MS 3.3 Students determine ΔS and ΔH from a graph of ΔG versus T.